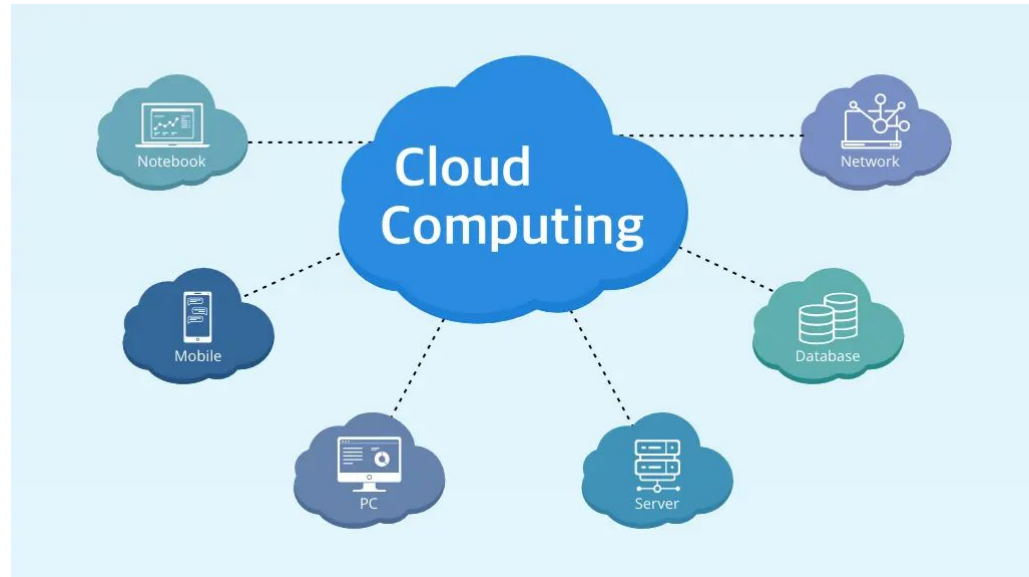


Cloud Foundations

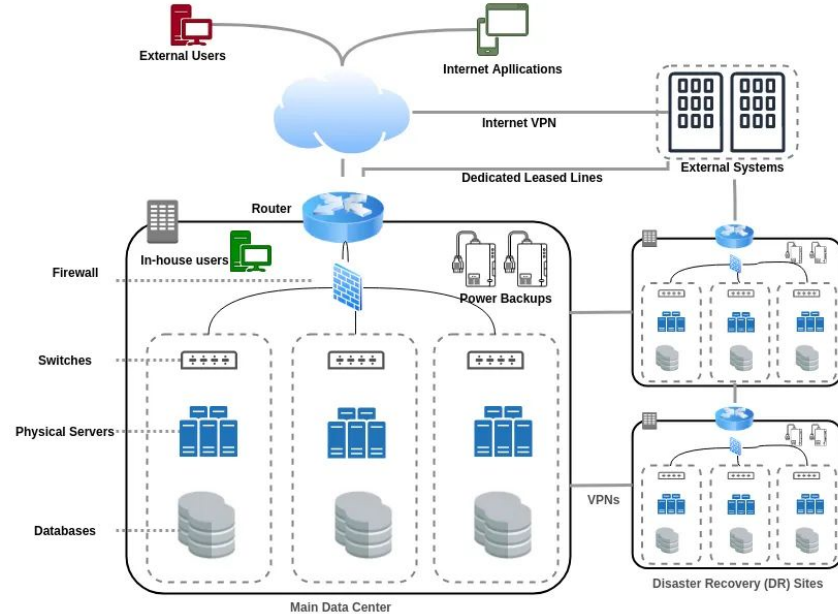
What is Cloud Computing?

Cloud computing is the delivery of computing services—including servers, storage, databases, networking, software, analytics, and intelligence—over the internet (“the cloud”) to offer faster innovation, flexible resources, and economies of scale. You typically pay only for cloud services you use, helping you lower your operating costs, run your infrastructure more efficiently, and scale as your business needs change.



World Before Cloud Computing

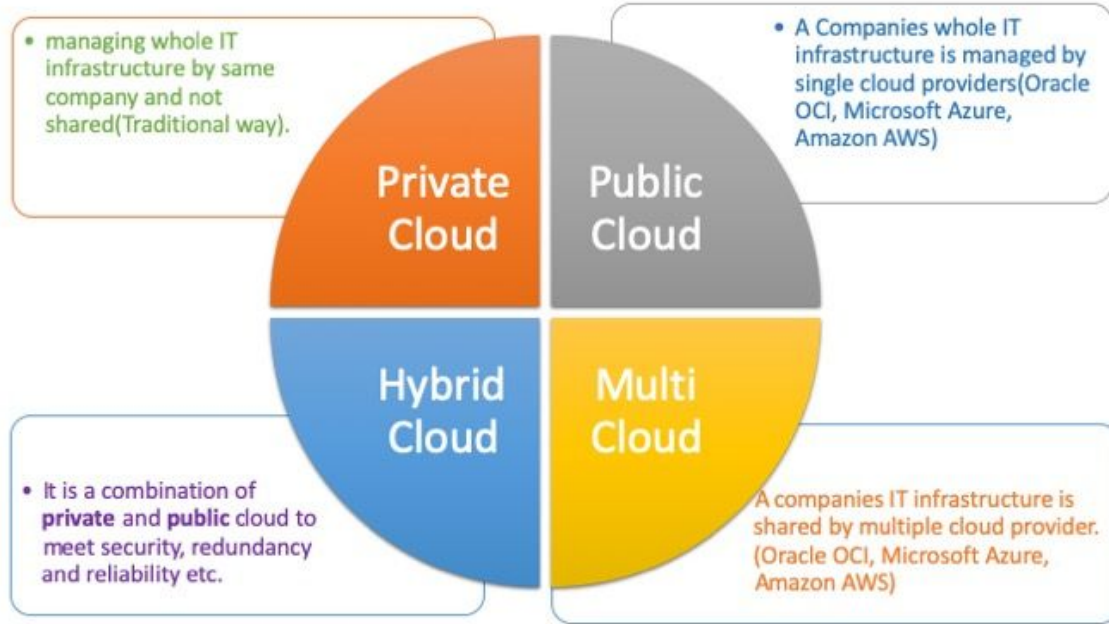
You own your data center!



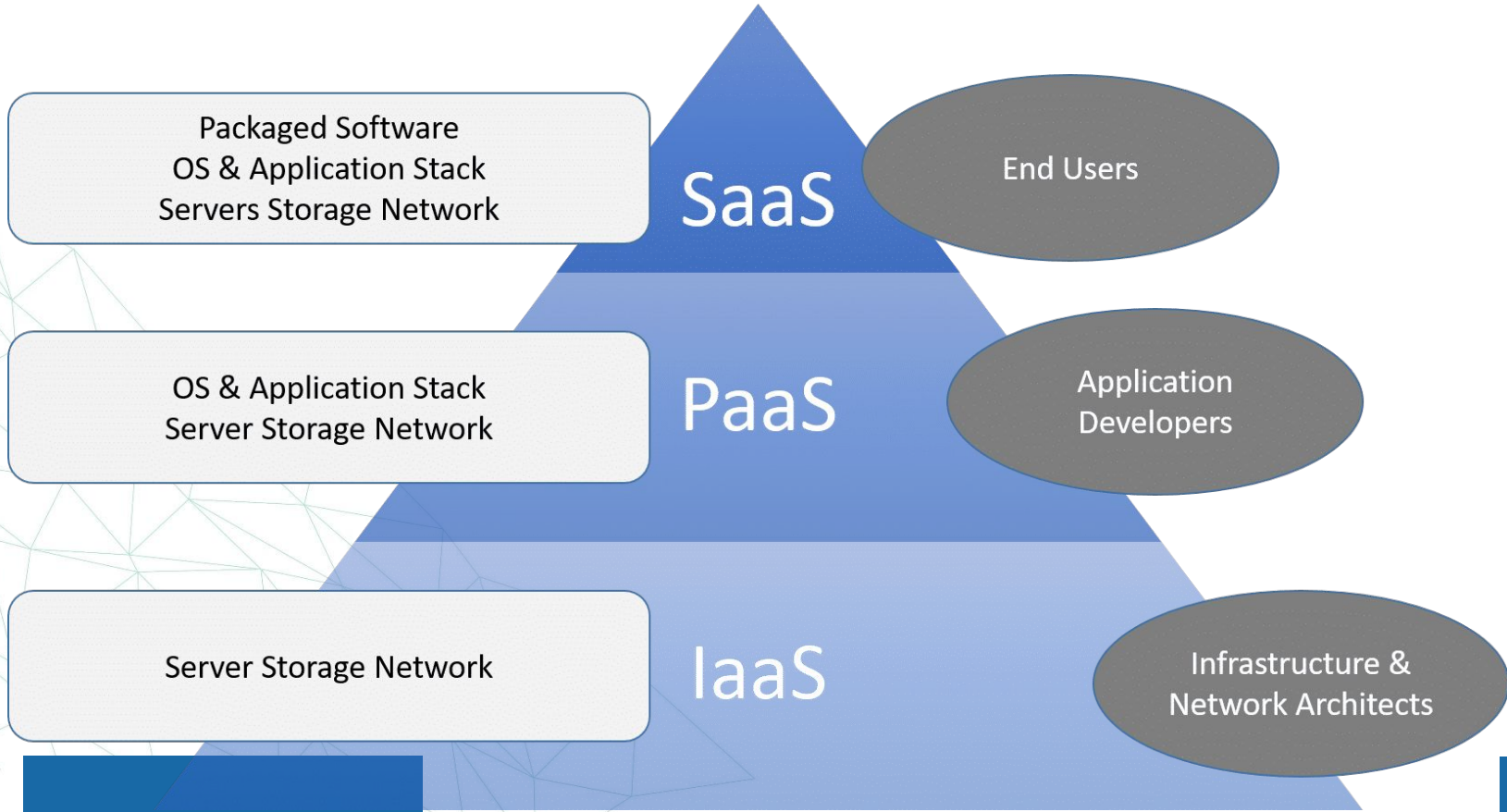
Top benefits of cloud computing

- Cost
- Speed
- Global
- Productivity
- Performance
- Reliability
- Security

Types of Cloud Computing



Cloud Service Models



Cloud Providers Comparison: Azure vs AWS vs GCP

Feature / Category	Microsoft Azure	Amazon Web Services (AWS)	Google Cloud Platform (GCP)
Market Share	2nd largest (strong enterprise presence)	Largest, market leader	3rd largest, growing rapidly
Strengths	Enterprise integration (esp. with MS Office, AD), Hybrid	Massive service catalog, maturity, ecosystem	Strong in analytics, AI/ML, Kubernetes
Compute Services	Azure VMs, Azure App Service, Azure Functions	EC2, Elastic Beanstalk, AWS Lambda	Compute Engine, App Engine, Cloud Functions
Storage Services	Blob Storage, Azure Files, Data Lake	S3, EFS, Glacier	Cloud Storage, Persistent Disks
Database Services	Azure SQL, Cosmos DB, PostgreSQL, MySQL	RDS, DynamoDB, Aurora	Cloud SQL, Firestore, Bigtable
Big Data & Analytics	Synapse, Data Factory, HDInsight	Redshift, Glue, EMR	BigQuery, Dataflow, Dataproc
AI/ML	Azure ML, Cognitive Services	SageMaker, Rekognition, Lex	Vertex AI, AutoML, Vision AI, Natural Language AI
CI/CD & DevOps	Azure DevOps, GitHub Actions (strong GitHub integration)	CodePipeline, CodeBuild, CodeDeploy	Cloud Build, Cloud Deploy
Kubernetes / Containers	AKS (Azure Kubernetes Service)	EKS (Elastic Kubernetes Service)	GKE (Google Kubernetes Engine)
Serverless	Azure Functions	AWS Lambda	Cloud Functions
Hybrid / Multi-cloud	Azure Arc, Azure Stack	Outposts, EKS Anywhere	Anthos
Security & Compliance	Azure Security Center, Sentinel	GuardDuty, Inspector, Macie	Security Command Center
Pricing	Pay-as-you-go, spot instances, cost calculator	Flexible, but can get complex	Simple pricing, per-second billing
Free Tier	12 months + always free services	12 months + always free tier	12 months + always free services
Learning Resources	Microsoft Learn, Azure Certifications	AWS Training, AWS Certifications	Google Cloud Skills Boost, GCP Certifications

Quiz

Q1:

"You only upload your code, and the cloud platform takes care of the rest – including servers and scaling."

What service model is this?

- A. IaaS
- B. PaaS
- C. SaaS
- D. On-Prem

Q2:

"You rent virtual machines, storage, and networking components and configure them as needed."
Which model does this describe?

- A. SaaS
- B. PaaS
- C. IaaS
- D. BaaS

Q3:

"You use a fully hosted email service like Microsoft Outlook 365 or Gmail."
What type of cloud model is being used?

- A. IaaS
- B. SaaS
- C. PaaS
- D. FaaS

Thank you